

Numerical Variables:

Frequency Distribution Tables and Histograms

A beginner-friendly guide, rebuilt from three Excel lessons (the frequency table lesson, its exercise, and the histogram lesson). No prior statistics needed.

What you will be able to do after this guide

1. Explain why a long list of numbers needs to be summarised.
2. Build a frequency distribution table by hand, step by step.
3. Read the table and say something useful about the data.
4. Turn the table into a histogram and read it.
5. Do the same in Excel with a few simple formulas.

The road map

Part	Question it answers	From workbook
1	Why do we need this at all?	Lesson 3
2	How do I build the table? (5 steps)	Lesson 3 and 5
3	What if a number lands exactly on an interval edge?	Exercise 4
4	What is a histogram and how do I read it?	Lesson 5
5	A full worked exercise	Exercise 4
6	Which Excel formulas do the work?	Lessons 3, 4, 5
7	Mistakes to avoid, and a practice problem	Summary

Words you will see (keep this table handy)

Word	Plain meaning	Example
Dataset	The list of numbers you collected	1, 9, 22, 24, ... 100
Interval (also called a bin)	A range of values that works like a bucket	21 to 41
Interval width	How wide each bucket is (all buckets share the same width)	20
Frequency	How many numbers fall in a bucket	4
Relative frequency	The same count as a share of the whole: frequency divided by total	$4 / 20 = 0.20$ (20%)
Frequency distribution table	The table listing every interval with its frequency	See Part 2
Histogram	That table drawn as bars	See Part 4

Part 1: Why do we need this?

Imagine a teacher gives you 20 test scores and asks, "How did the class do?" Here they are:

1, 9, 22, 24, 32, 33, 42, 44, 57, 66, 70, 73, 75, 76, 79, 82, 87, 89, 95, 100

Looking at that list, can you tell whether most students scored high or low? It is hard. Your eyes see twenty separate numbers, not a picture.

First attempt: count each number on its own. This is what the first column of the lesson workbook does. The result: every number appears exactly **once**, so every count is 1. A column of twenty 1s tells you nothing.

The fix: group the numbers into buckets (intervals) and count how many land in each bucket. Now patterns show up. That grouped table is called a **frequency distribution table**.

Quick link to what you already know

For **categories** (Audi, BMW, Mercedes) you count each category directly, because there are only a few. For **numbers** that can take many values (scores, prices, weights) you first create intervals, then count. That is the only real difference.

Part 2: Build the table in 5 steps

We will use the same 20 test scores. Follow the steps in order; each one feeds the next.

Step 1: Find the smallest and the largest number

Smallest = **1**, largest = **100**. These two numbers decide how far the buckets must reach.

Step 2: Decide how many intervals you want

There is no strict rule. Too few intervals hides detail; too many makes the table messy again. For a small dataset, 5 to 10 is common. The lesson picks **5**.

Step 3: Work out the interval width

Spread the full range evenly over your intervals:

Interval width = (largest - smallest) / number of intervals = $(100 - 1) / 5 = 19.8$

19.8 is awkward, so we **round up** to **20**. Round *up*, not down: with a width of 19, five intervals would only reach $1 + 5 \times 19 = 96$, and the number 100 would be left out.

Step 4: Write the intervals

Start at the smallest number (1). Add the width to get the end ($1 + 20 = 21$). The next interval starts where the last one ended. Repeat until you have 5:

1 to 21, 21 to 41, 41 to 61, 61 to 81, 81 to 101

The last one ends at 101, not 100. That tiny extra room comes from rounding the width up, and it is fine.

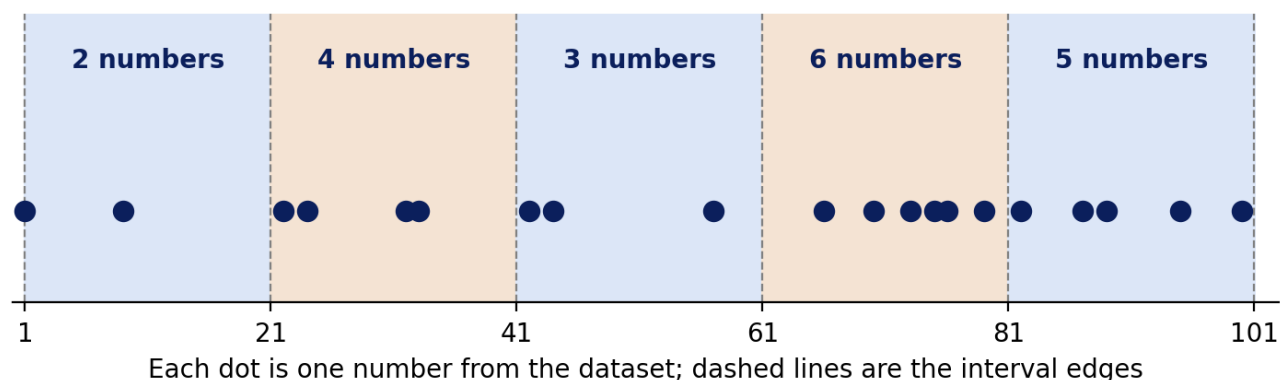


Figure 1. The 20 numbers placed on a number line, with the five intervals shaded. Counting the dots in each shaded area gives the frequencies.

Step 5: Count the numbers in each interval

Go through the dataset and count. Then divide each count by the total (20) to get the relative frequency.

Interval	Numbers inside it	Frequency	Relative frequency
1 to 21	1, 9	2	$2 / 20 = 0.10$
21 to 41	22, 24, 32, 33	4	$4 / 20 = 0.20$
41 to 61	42, 44, 57	3	$3 / 20 = 0.15$
61 to 81	66, 70, 73, 75, 76, 79	6	$6 / 20 = 0.30$
81 to 101	82, 87, 89, 95, 100	5	$5 / 20 = 0.25$
Total		20	1.00

Always do this check

The frequencies must add up to the number of values in your dataset (here 20), and the relative frequencies must add up to 1.00. If not, you missed or double-counted a number.

What does the table tell us? 30% of the scores (6 out of 20) are between 61 and 81, the most crowded interval. Only 10% are below 21. So most students scored in the upper half. The raw list never made that obvious.

Part 3: What if a number lands exactly on an edge?

Look at the numbers 21, 41, 61 and 81. Each one is the **end** of one interval and the **start** of the next. If a score were exactly 21, which interval would it go in? It must be counted only once, so we need a rule.

The rule used in these lessons: the first interval includes both of its ends. Every other interval leaves out its start and includes its end. It is written with brackets:

Bracket	Meaning	Example
[] square	The number IS included	[1,21] includes both 1 and 21
() round	The number is NOT included	(21,41] means more than 21, up to and including 41

So our five intervals are written **[1,21], (21,41], (41,61], (61,81], (81,101]**. Every number belongs to exactly one of them. A score of 21 goes in the first interval; a score of 22 goes in the second.

Heads-up: an edge case hiding in the lesson files

The dataset in lesson 3 has the number **41**, while lesson 5 has **42** in the same place. With 42, nothing sits on an edge, so every method agrees (2, 4, 3, 6, 5). With 41, it depends on the rule: the bracket rule above puts 41 in (21,41], while lesson 3's Excel formula counts it in the 41-to-61 interval.

Neither is "wrong". The lesson to take away: **pick one rule and use it for every interval**. That is why the course writes the brackets down.

Part 4: The histogram

A **histogram** is the frequency table drawn as bars. Put the intervals on the horizontal axis, and make each bar as tall as its frequency. That is all it is.

Two things make it different from the bar chart you saw for car brands:

	Bar chart	Histogram
Used for	Categories (Audi, BMW, ...)	Numbers grouped into intervals
Bars	Separated by gaps	Touching, because one interval ends exactly where the next begins
Order of bars	Can be rearranged	Fixed: left to right along the number line

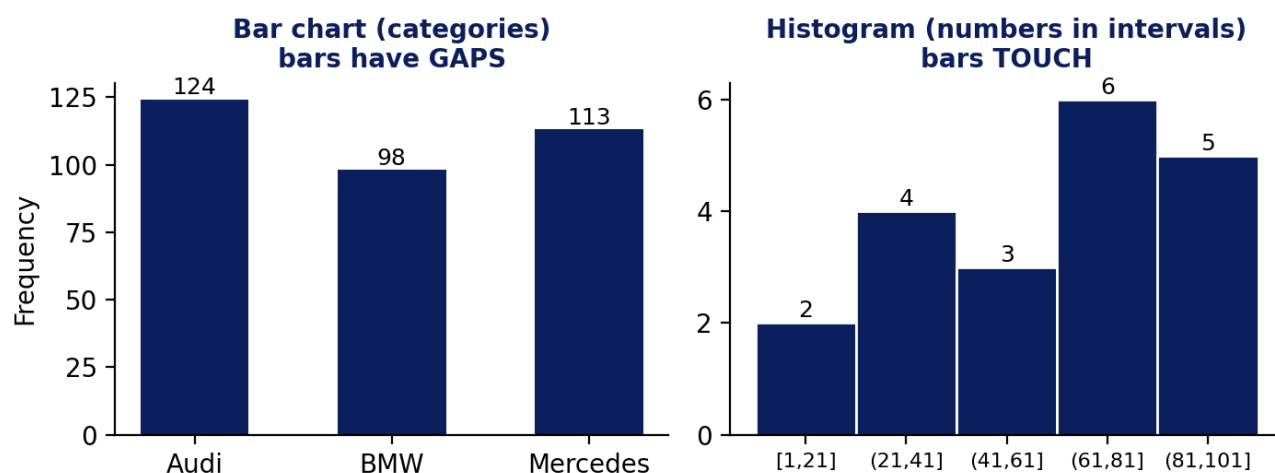


Figure 2. Same idea, two chart types. Gaps for categories, no gaps for numeric intervals.

How to read ours: the tallest bar (61 to 81) is where the scores crowd. The bars climb from left to right and then dip at the end, so more students scored high than low.

Relative frequency histogram. You can also draw the bars using the relative frequencies. The shape is identical; only the vertical scale changes, from counts (2, 4, 3, 6, 5) to shares (0.10, 0.20, 0.15, 0.30, 0.25). Use shares when you want to compare datasets of different sizes.

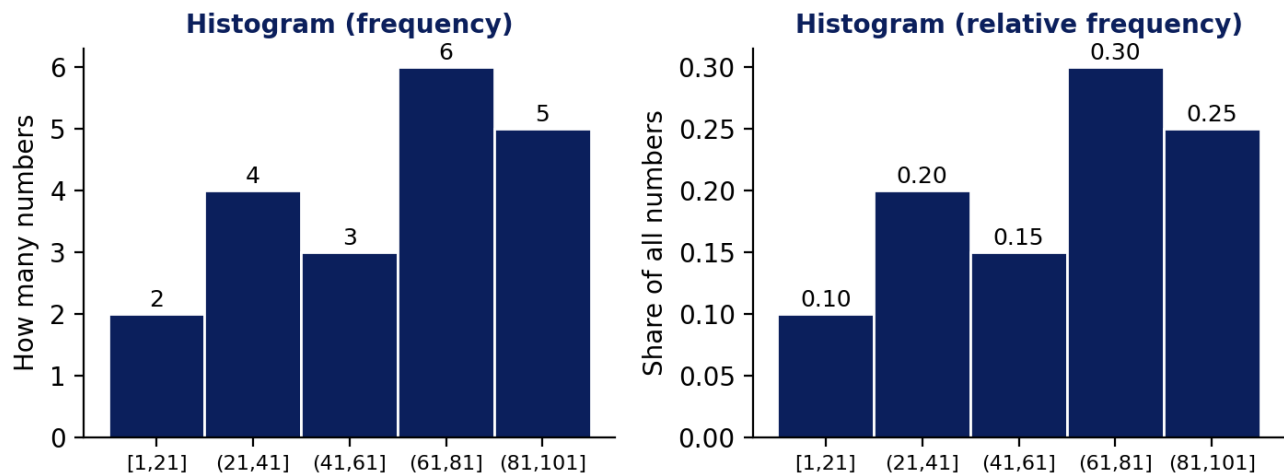


Figure 3. Left: frequency. Right: relative frequency. Same shape.

Making a histogram in Excel

1. Select your data. 2. Insert, then Charts, then Histogram. 3. To control the intervals: click the horizontal axis, open Format Axis (Axis options), and set the bin width or the number of bins. If you skip step 3, Excel chooses its own bins and your chart will not match your table.

Part 5: A full worked exercise

This is the exercise workbook, solved the same 5-step way. Here is the dataset (20 numbers, already in order):

8, 30, 30, 50, 86, 94, 102, 110, 169, 170, 176, 236, 240, 241, 242, 255, 262, 276, 279, 282

Your tasks: split the numbers into 6 intervals of equal width, find the width, and build the frequency table. Then repeat with the exact width, without rounding up.

Task 1: the interval width

Smallest = 8, largest = 282. Width = $(282 - 8) / 6 = 274 / 6 = 45.67$. Round up to the next whole number: **46**.

Task 2: the table (using width 46)

Start at 8: $8 + 46 = 54$. Next interval starts at 54: $54 + 46 = 100$. Keep going. Then count, using the bracket rule from Part 3.

Interval	Numbers inside it	Frequency	Relative frequency
[8, 54]	8, 30, 30, 50	4	0.20
(54, 100]	86, 94	2	0.10
(100, 146]	102, 110	2	0.10
(146, 192]	169, 170, 176	3	0.15
(192, 238]	236	1	0.05
(238, 284]	240, 241, 242, 255, 262, 276, 279, 282	8	0.40
Total		20	1.00

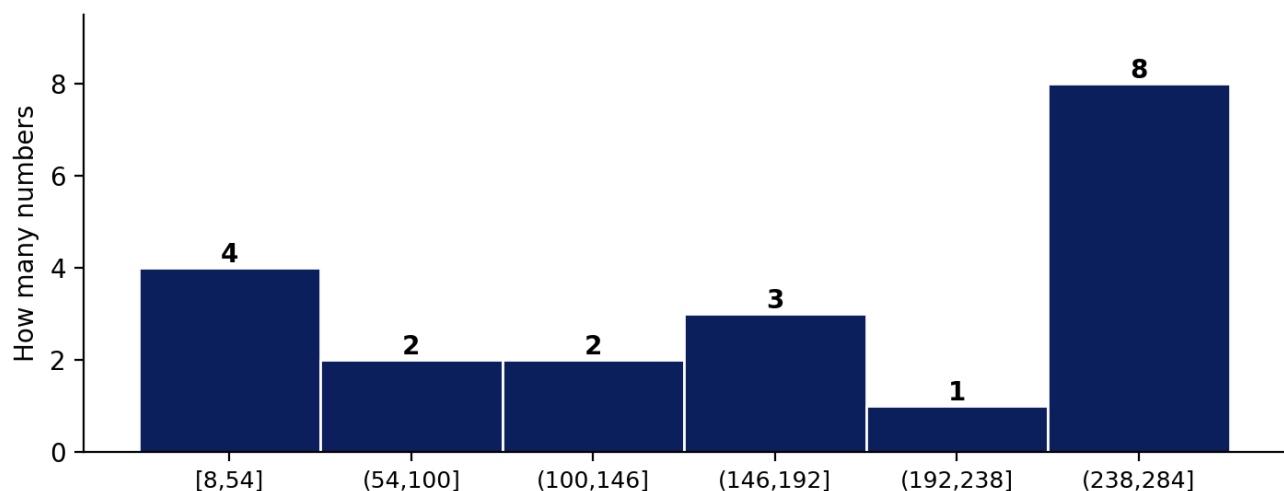


Figure 4. Histogram of the exercise data (width 46).

What does it tell us? The data is not crowded in the middle. There is a small group at the low end (4 numbers up to 54) and a big group at the top end: 8 of the 20 numbers, which is **40%**, fall in the last interval. The middle is thin.

Task 3: repeat with the exact width (45.67)

Now do not round. Each interval is exactly 45.67 wide, so the edges are 8, 53.67, 99.33, 145, 190.67, 236.33 and 282.

Rounded width (46)	Exact width (45.67)	Frequency	Relative frequency
[8, 54]	[8, 53.67]	4	0.20
(54, 100]	(53.67, 99.33]	2	0.10
(100, 146]	(99.33, 145]	2	0.10
(146, 192]	(145, 190.67]	3	0.15
(192, 238]	(190.67, 236.33]	1	0.05
(238, 284]	(236.33, 282]	8	0.40

Why are the counts identical? The edges only moved by a fraction (54 became 53.67, and so on), and no number in this dataset sits in that tiny gap. So no number changes bucket.

Which should you use? The exact width is the norm: the first interval starts exactly at the smallest number (8) and the last ends exactly at the largest (282). The rounded width is easier to read, and for most datasets gives the same answer.

Part 6: The Excel formulas, in plain English

Suppose your numbers are in cells B4 to B23, your first interval starts in E9 and ends in F9, and the number of intervals is in F5. Here is what each formula does.

What you need	Formula	In plain English
Smallest / largest	=MIN(B4:B23) =MAX(B4:B23)	Finds the lowest and highest number
Interval width (rounded up)	=ROUNDUP((MAX(B4:B23)-MIN(B4:B23))/F5,0)	(largest - smallest) / intervals, then round UP to a whole number. The 0 means zero decimal places
Interval width (exact)	=(MAX(B4:B23)-MIN(B4:B23))/F5	The same, with no rounding
Interval end	=E9+\$F\$6	Start + width. The next interval's start is simply the previous end
Frequency (first interval)	=COUNTIF(B4:B23,">="&E9)-COUNTIF(B4:B23,">"&F9)	Numbers at or above the start, minus numbers above the end = numbers inside the interval, edges included
Frequency (other intervals)	=COUNTIF(B4:B23,">"&E10)-COUNTIF(B4:B23,">"&F10)	Same idea, but the start is excluded, matching the bracket rule
Relative frequency	=G9/COUNT(B4:B23)	Frequency divided by how many numbers there are
Total check	=SUM(G9:G13)	Should equal the number of values (20)

How the COUNTIF trick works

COUNTIF counts the cells that meet a condition. In the 20 test scores, 18 numbers are above 21 and 14 are above 41. Then $18 - 14 = 4$ numbers lie above 21 but not above 41, which is exactly the interval (21, 41]. Subtracting two counts is a quick way to count "between".

Part 7: Mistakes to avoid

Mistake	Why it hurts	Fix
Rounding the width down	The last number falls outside every interval	Round UP
Intervals of different widths	The histogram bars become misleading	Use one width for all intervals
Gaps or overlaps between intervals	A number is lost, or counted twice	Start each interval where the last ended and use the bracket rule
Totals do not add up	You miscounted	Frequencies must total N, relative frequencies must total 1.00
Gaps between histogram bars	It becomes a bar chart, which is for categories	Make the bars touch
Too few or too many intervals	Detail is hidden, or the table is as messy as the raw data	Try 5 to 10 for small datasets

Part 8: Try it yourself

Cover the answer and try first.

Practice problem

The ages of 12 people at a workshop are:

18, 22, 25, 27, 31, 34, 35, 38, 42, 45, 51, 58

Build a frequency distribution table with **4 intervals**: find the width, write the intervals, and count the frequencies and relative frequencies.

Answer

Width = $(58 - 18) / 4 = 40 / 4 = \mathbf{10}$, a whole number, so there is nothing to round. Intervals start at 18: 18 to 28, 28 to 38, 38 to 48, 48 to 58.

Interval	Ages inside it	Frequency	Relative frequency
[18, 28]	18, 22, 25, 27	4	$4 / 12 = 0.33$
(28, 38]	31, 34, 35, 38	4	$4 / 12 = 0.33$
(38, 48]	42, 45	2	$2 / 12 = 0.17$
(48, 58]	51, 58	2	$2 / 12 = 0.17$
Total		12	1.00

Notice that 38 sits exactly on an edge. By the bracket rule it belongs to (28, 38], not to the next interval. A third of the group is under 28, and another third is between 29 and 38.

One-page summary

The whole topic in 6 lines

1. A long list of numbers is hard to read, so group them into **equal-width intervals** and count.
2. **Width** = (largest - smallest) / number of intervals, rounded up (or kept exact).
3. Start at the smallest number; each interval starts where the last one ended.
4. **Frequency** = how many numbers in the interval. **Relative frequency** = frequency / total.
5. Frequencies must add up to the total number of values; relative frequencies to 1.00.
6. A **histogram** is the table drawn as touching bars: tall bar = crowded interval.